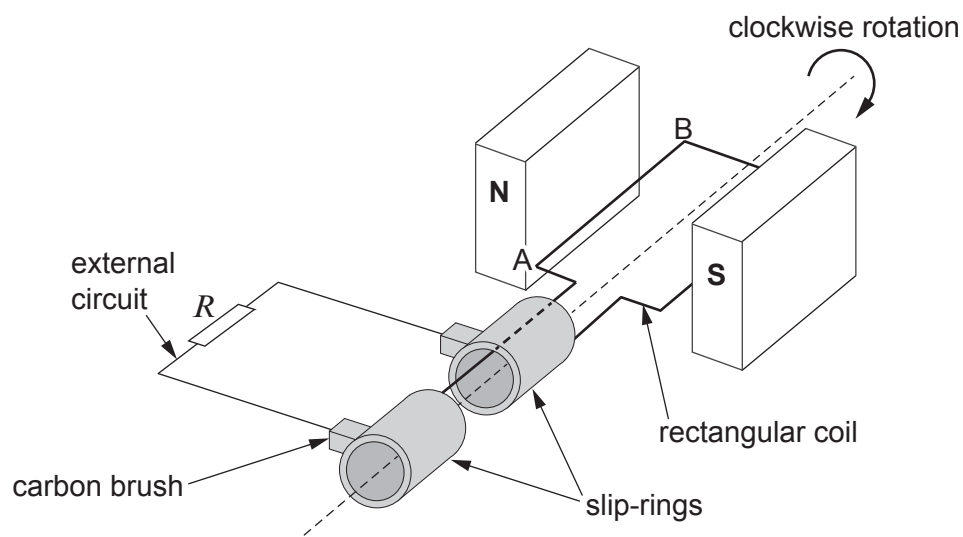


9. Electromagnetic induction explains the operation of electric generators and transformers.

(a) The diagram shows an a.c. generator.



(i) Explain how Fleming's right hand rule is used to determine the direction of the induced current in the side labelled AB **and** state this direction. [3]

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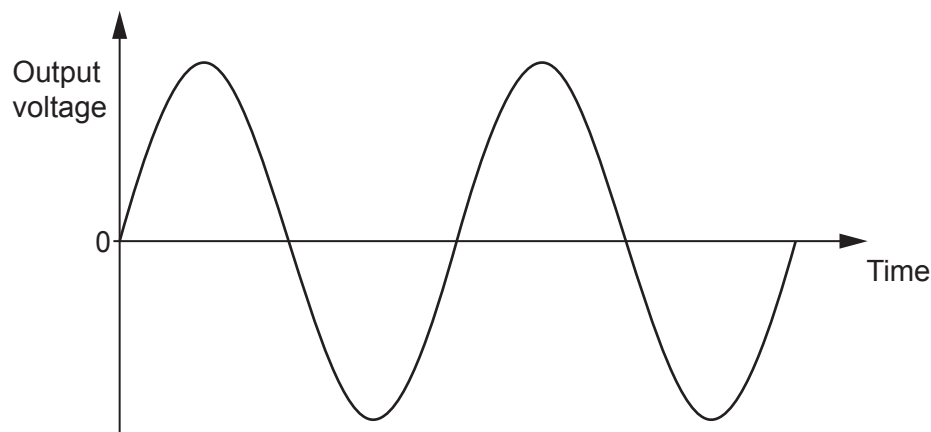
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(ii) The output from an a.c. generator is shown in the graph below.

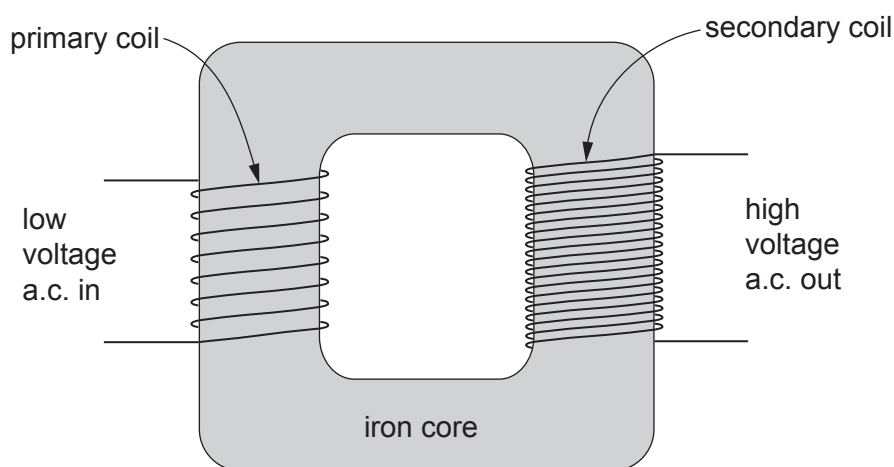


Three changes that could be made to a generator are listed in the first column of the table below.
Complete the table by placing **two** ticks (✓) in each row. [3]

Change	Effect on voltage			Effect on time for 1 cycle		
	Decreases	Stays the same	Increases	Decreases	Stays the same	Increases
Weaker magnets						
Coil spins faster						
More turns in the coil						



- (b) The diagram shows a step-up transformer used in the National Grid.



- (i) Explain how an alternating voltage applied to the primary coil produces an alternating voltage across the secondary coil. [2]

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- (ii) A transformer increases voltage from 25 000 V to 400 000 V. The primary coil contains 600 turns of wire. Use an equation from page 2 to calculate the number of turns in the secondary coil, N_2 . [3]

$N_2 =$

